

permutations of {A, B, C} ranging from AAA to CCC, but you could also, for example, have five-letter permutations with repetition ranging from AAAAA to CCCCC. The number of permutations with repetition of n elements that are k elements long is n^k . [Table 6-2](#) lists it as n^n for permutations with repetition that are also n elements long.

Ordering matters for permutations, but not for combinations. While AAB, ABA, and BAA are considered the same combination with repetition, they are considered three separate permutations with repetition.